

MISSION 100

JEE 2027

Lecture – 01

Mathematics

BASIC MATHEMATICS

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Physics





Topics to be covered

1

Solving Inequality

2

Wavy Curve Method

3

Modulus Equation

4

Modulus Inequality

Red {

- Attend lec.
- Short & crisp Notes.
- Re-attempt all examples of class.

Yellow {

- QPP
- HW (Module Quesⁿ.)

Green {

- PYQ's (50 per chapter)
- Test
- Weekly Revision.

INEQUALITY



GOLDEN RULES OF INEQUALITY

✓ **Rule-01:** We can add (or subtract) any number ' k ' on both sides of inequality. Doing this will not change the sign of inequality.

✓ **Rule-02:**

We can multiply (or divide) any non-zero number ' k ' on both sides of inequality and sign of inequality will change according to sign of ' k ' as:

If $k > 0$, then sign of inequality will remains same,

If $k < 0$, then sign of inequality will get reversed.



SOLUTION REPRESENTATION

Column-I

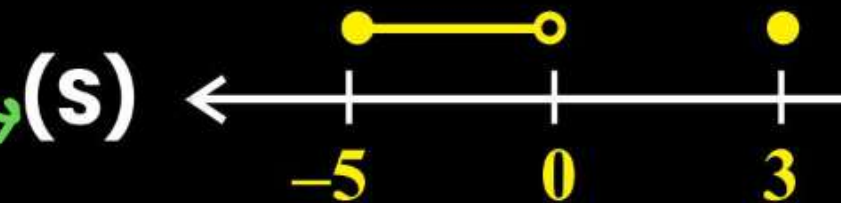
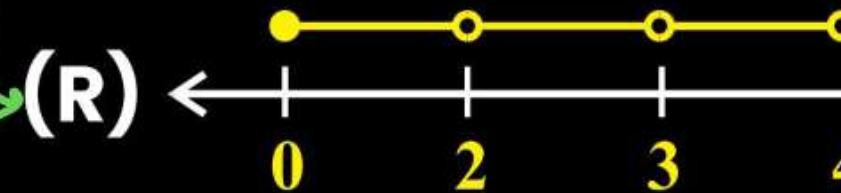
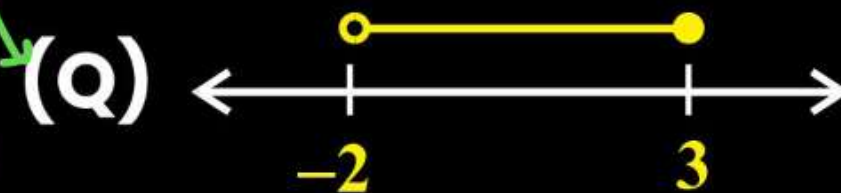
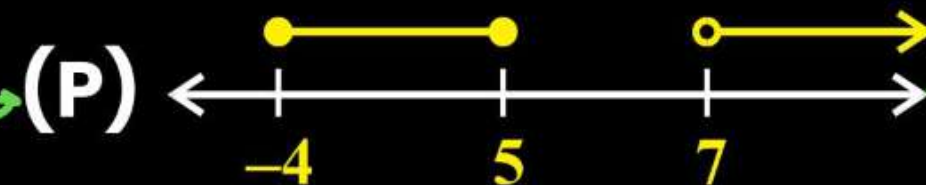
(A) $-2 < x \leq 3, x \in R$

(B) $0 \leq x < 4, x \in R \text{ \& } x \neq 2, 3$

(C) $-5 \leq x < 0, x \in R \text{ or } x = 3$

(D) $-4 \leq x \leq 5 \text{ or } x > 7$

Column-II



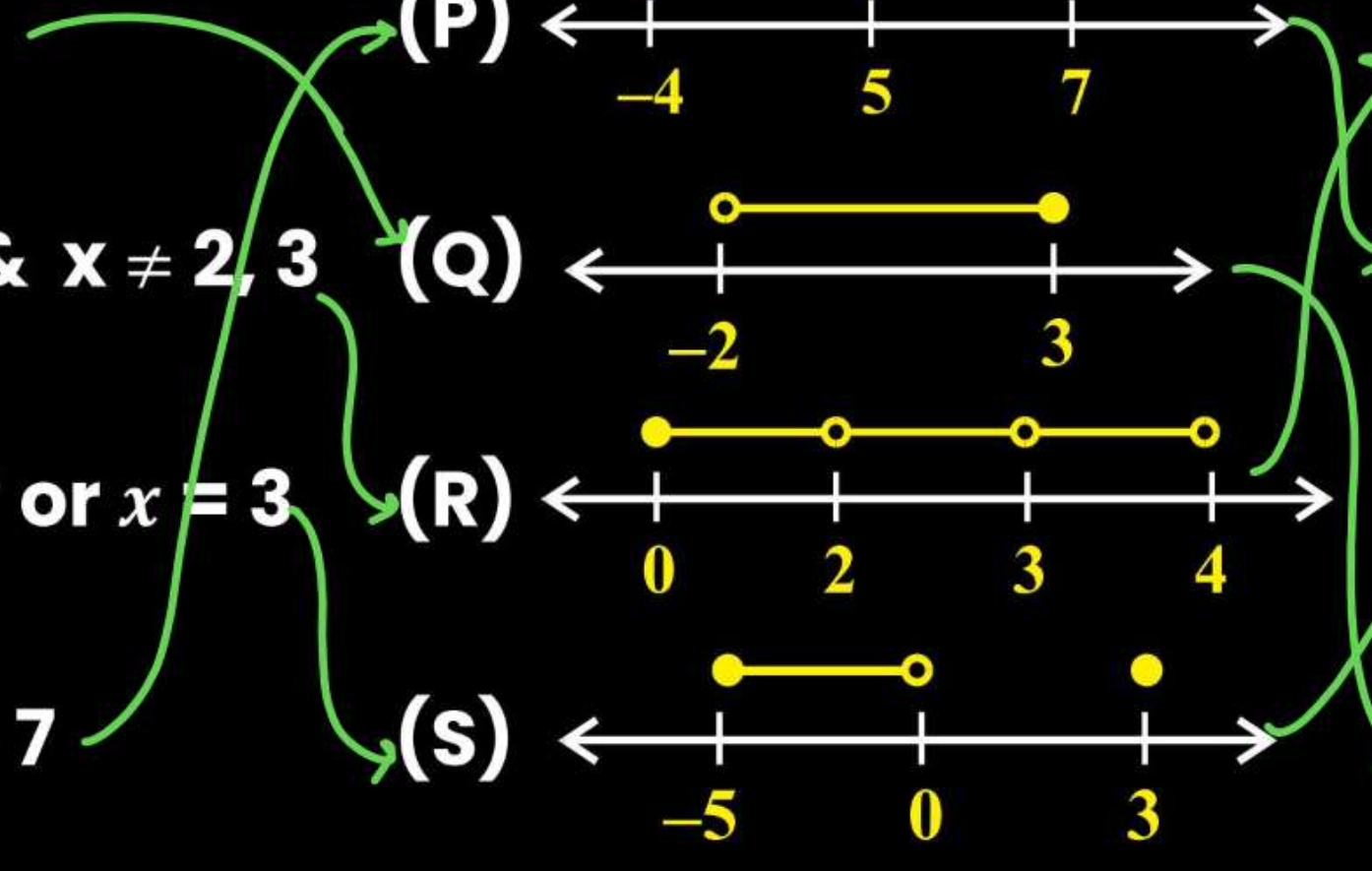
Column-III

(X) $x \in [0, 4) - \{2, 3\}$

(Y) $x \in [-4, 5] \cup (7, \infty)$

(Z) $x \in [-5, 0) \cup \{3\}$

(T) $x \in (-2, 3]$





WAVY CURVE METHOD



Step-01: Make RHS zero.

Step-02: Convert Numerator & Denominator of LHS into Linear Factors

Step-03: Where each Linear Factor becomes Zero is called Critical Point (CP), plot all critical points on Number Line

Step-04: Check sign of given expression by putting values of x from different parts of number line.

Step-05: Jo pucha gya, use dekh key jawab do.

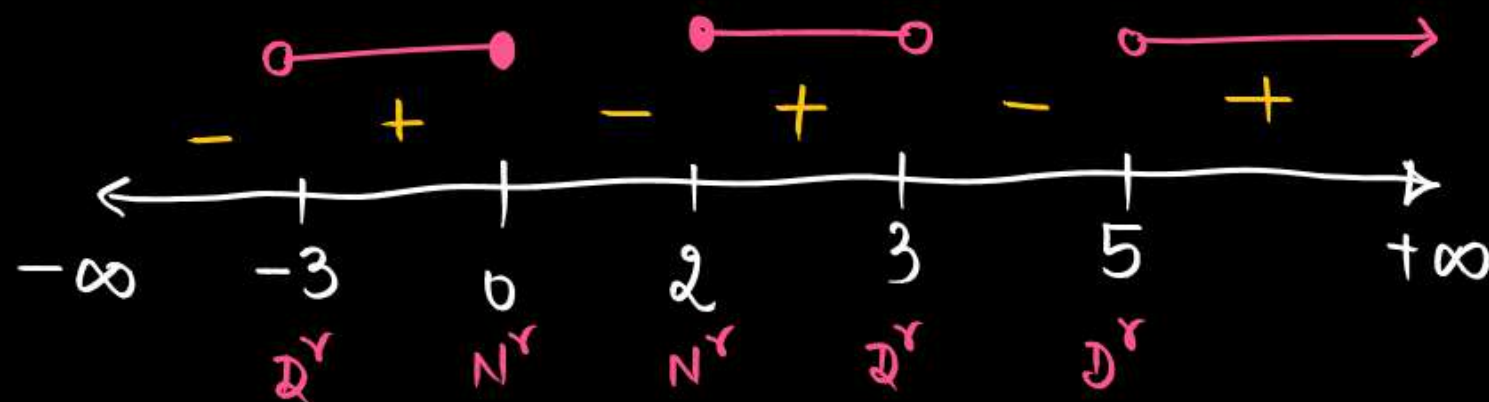
Question

Find exhaustive(complete) set of values of x satisfying:

$$\frac{x(x-2)}{(x-5)(x^2-9)} \geq 0$$

$$x^2 - 3^2 = (x-3)(x+3)$$

$$\frac{x(x-2)}{(x-5)(x-3)(x+3)} \geq 0$$



$$x \in (-3, 0] \cup [2, 3) \cup (5, \infty)$$



ALWAYS REMEMBER



1. **Roots of denominator always** → *Hollow Circle*.
2. **In case of $\geq, \leq$ roots of numerator** → *Dark Circle*.
3. **In case of $>, <$ all roots** → *Hollow Circle*.

The set of real values of x for which $\frac{10x^2+17x-34}{x^2+2x-3} < 8$, is

A $\left(-\frac{5}{2}, 2\right)$

☒ **B** $\left(-3, -\frac{5}{2}\right) \cup (1, 2)$

C $(-3, 1)$

D $(-3, 2)$

$$\frac{10x^2+17x-34}{x^2+2x-3} - 8 < 0$$

$$\frac{10x^2+17x-34-8x^2-16x+24}{x^2+2x-3} < 0$$

$$x^2+2x-3$$

$\swarrow \quad \searrow$
 $3x \quad -x$

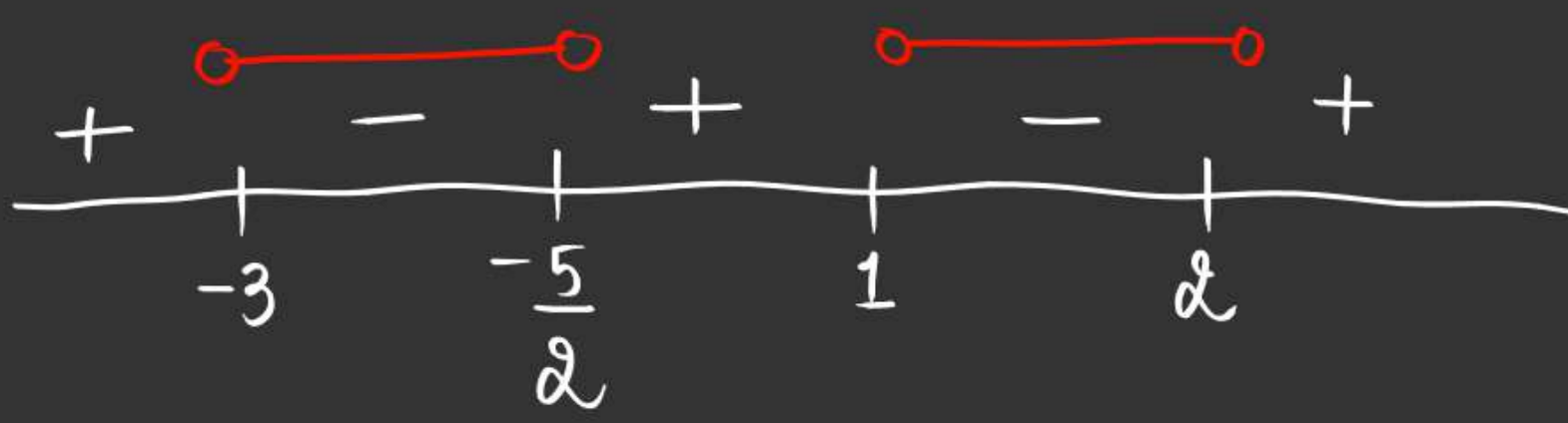
$$\frac{2x^2+x-10}{(x-1)(x+3)} < 0$$

$$\frac{(x-2)(2x+5)}{(x-1)(x+3)} < 0$$

$$\frac{(x-2)(2x+5)}{(x-1)(x+3)} < 0$$

2 (points to $x-2$)
 $-\frac{5}{2}$ (points to $2x+5$)
 $\underline{1}$ (points to $x-1$)
 -3 (points to $x+3$)

<



Find the set of all x for which $\frac{2x}{2x^2+5x+2} > \frac{1}{x+1}$.

$$\frac{2x}{2x^2+5x+2} - \frac{1}{x+1} > 0$$

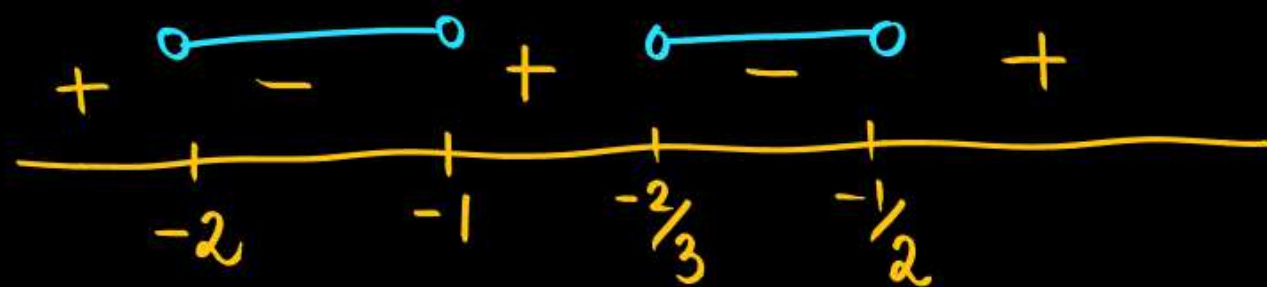
$$\frac{\cancel{2x^2} + 2x - \cancel{2x^2} - 5x - 2}{(2x^2+5x+2)(x+1)} > 0$$

$$2x^2 + 4x + x + 2$$

$$2x(x+2) + 1(x+2)$$

$$\frac{(3x+2)}{(2x+1)(x+2)(x+1)} < 0$$

$\begin{matrix} -\frac{2}{3} \\ -\frac{1}{2} & -2 & -1 \end{matrix}$



$$x \in (-2, -1) \cup \left(-\frac{2}{3}, -\frac{1}{2}\right)$$



TRICK



1. Make coefficient ' x ', **positive** in each linear factor.
2. Now start marking $\rightarrow (+ / - / + / - / + / -)$ from right to left.
(po ne
3. **Remember :**



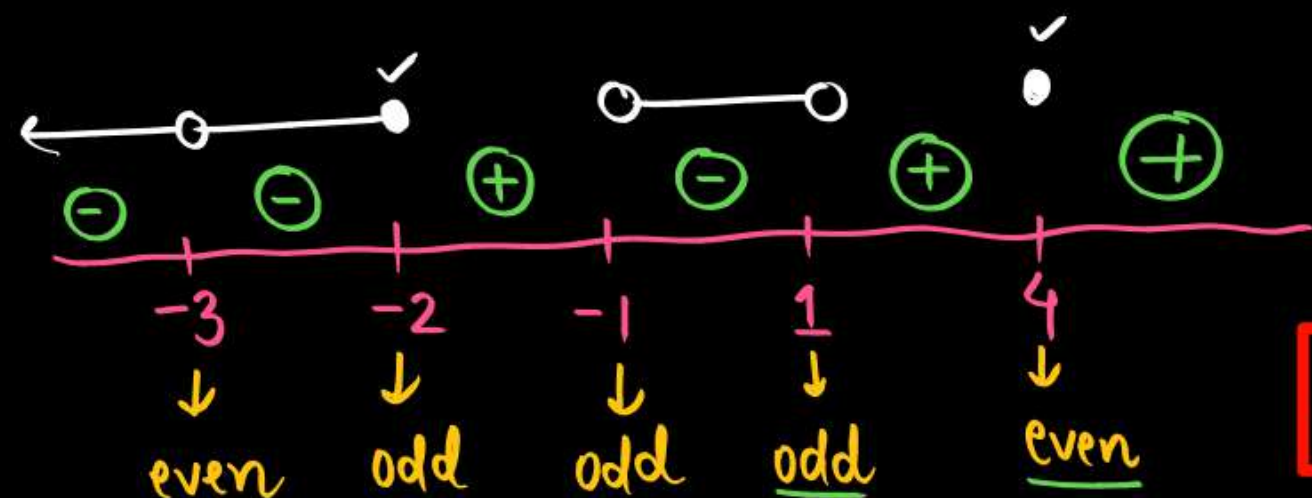
Jinn factors ki **net power odd hai**, uske about $\rightarrow$ sign change hoga.

Jinn factors ki **net power even hai**, uske about $\rightarrow$ sign will not change.

Question

Answer each of following part using trick :

(i)
$$\frac{(x+2)^1(x-4)^2}{(x+1)^1(x+3)^4(x-1)^3} \leq 0$$



$$x \in (-\infty, -3) \cup (-3, -2] \cup (-1, 1) \cup \{4\}$$

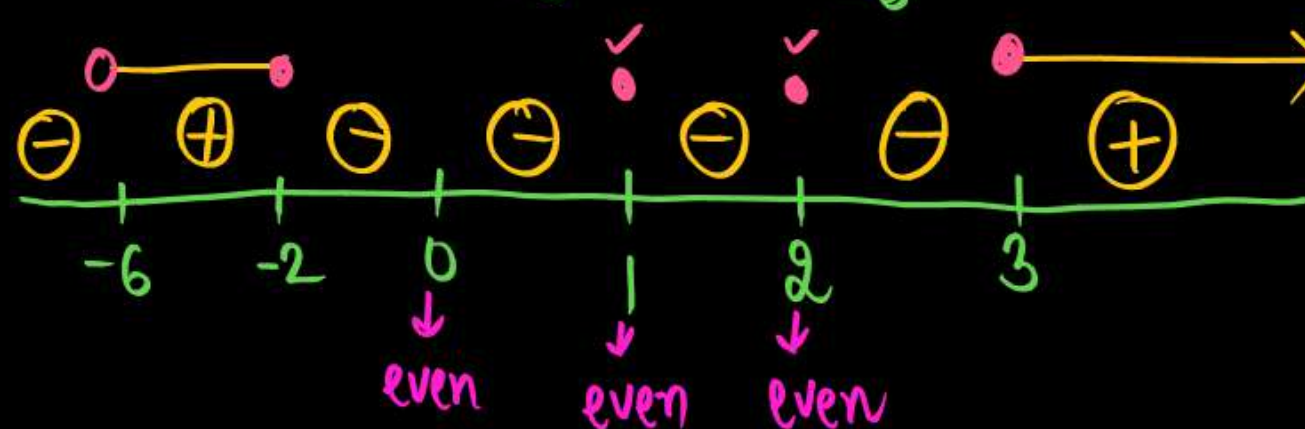
(ii)

$$\frac{(x-1)^2(x^2-4)(x^2-5x+6)}{(x^3+6x^2)} \geq 0$$

$$x^2(x+6)$$

$$\frac{(x-1)^2(x-2)^2(x+2)(x-3)}{x^2(x+6)} \geq 0$$

$$x \in (-6, -2] \cup [3, \infty) \cup \{1, 2\}$$



Question

$$\frac{a^m}{a^n} = a^{m-n}$$



Solution set of inequality $\left(\frac{x^2-19x+88}{x^2-11x+24}\right) \leq 0$ is

- A** $x \in (3, 11]$
- B** $x \in [3, 11]$
- C** $x \in (3, 11] - \{8\}$
- D** $x \in (3, 11) - \{8\}$

Handwritten solution steps:

$$\frac{(x-11)^1 (x-8)^1}{(x-3)^1 (x-8)^1} \leq 0$$

Canceling $(x-8)^1$ (noting $(x-8)^0 = 1$):

$$\frac{x-11}{x-3} \leq 0 \quad \left\{ \begin{array}{l} x-8 \neq 0 \\ x \neq 8 \end{array} \right.$$

Sign chart for $\frac{x-11}{x-3} \leq 0$:

Number line with critical points at 3 and 11. Sign is positive for $x < 3$, negative for $3 < x < 11$, and positive for $x > 11$. The inequality is satisfied in the interval $(3, 11)$.

Sign chart for the original inequality (including $x=8$):

Number line with critical points at 3, 8, and 11. Sign is positive for $x < 3$, negative for $3 < x < 8$, negative for $8 < x < 11$, and positive for $x > 11$. The inequality is satisfied in the interval $(3, 11]$, but $x=8$ is excluded because it makes the denominator zero.

Final solution set: $x \in (3, 11] - \{8\}$

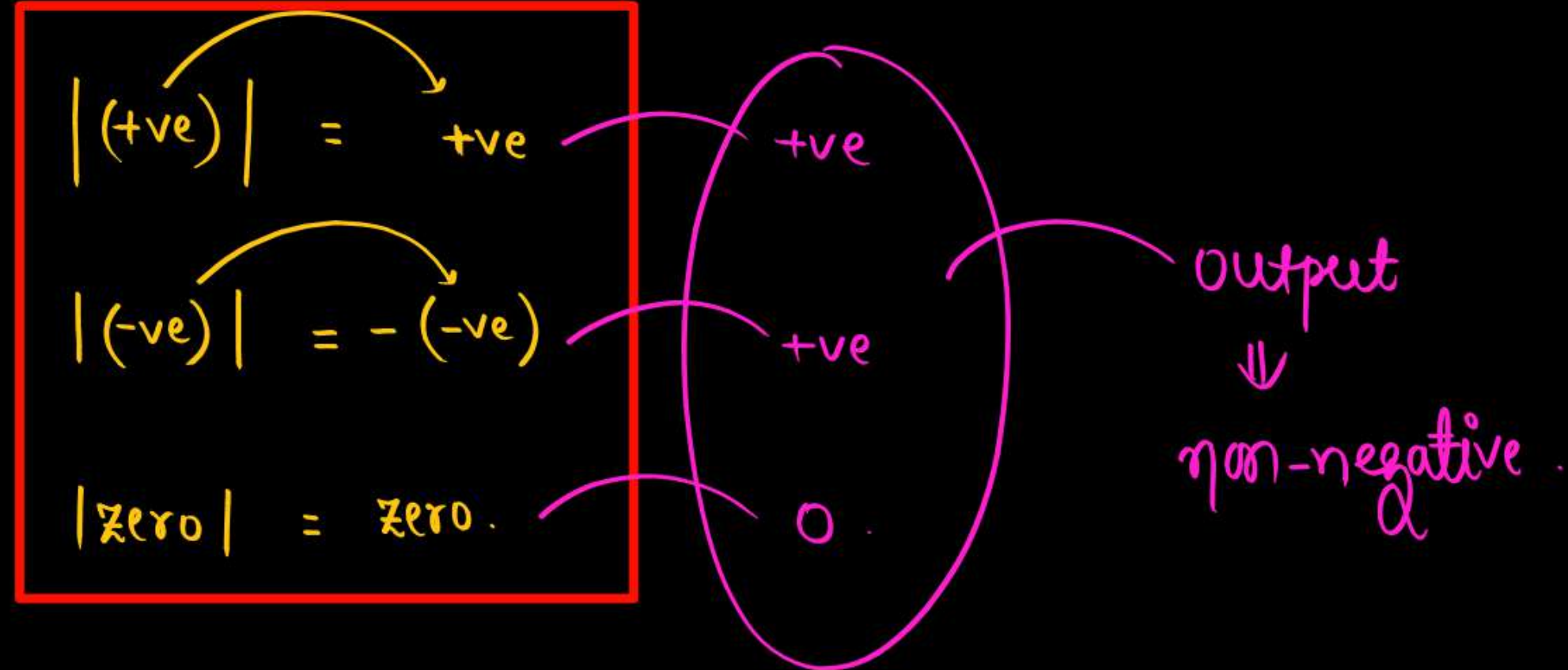
MODULUS



MODULUS FUNCTION



Also called as: **Absolute value Function**



RAPID FIRE ROUND



(1) $|-11| = \underline{-(-11)} = \underline{11} \checkmark$

(2) $|5 - \sqrt{41}| = \underline{-(5 - \sqrt{41})} = \sqrt{41} - 5$

(3) $|-13| = |13|$ (True / False)

(4) If $x > 0$ then $\frac{|x|}{x} = \frac{x}{x} = \underline{1}$.

(5) If $x < 0$ then $\frac{|x|}{x} = \frac{-x}{x} = \underline{-1}$.

(6) Value of x satisfying: $|x| = -2$ is $\underline{\emptyset}$.

(7) Value of Mod is non-negative (True/False)

(8) $|0| = \underline{0}$.



MODULUS EQUATION

RAPID FIRE ROUND

100%
Quesⁿ.

Solve following equations:

(i) $|x| = 7$ $\rightarrow x = \pm 7$

(ii) $|x| = -4$ $\rightarrow \phi$

(iii) $|x + 5| = 0$ $\rightarrow \phi$

(iv) $|x - 2| = 6$ $\rightarrow x + 5 = 0$
 $x = -5$

$|x - 2| = 6$
 $x - 2 = 6, -6$
 $x = 8, -4$

Note:-

$|x| = \alpha$
 $\rightarrow +ve$

Solⁿ
 $x = \pm \alpha$

$$x = 12, -8.$$

$$\# \quad \overset{x}{\left| |x-2| - 3 \right|} = 7.$$

$$|x| = 7$$

$$\begin{aligned} & \swarrow x = 7 \\ & |x-2| - 3 = 7 \end{aligned}$$

$$|x-2| = 10$$

$$\text{OR} \quad \begin{aligned} & \swarrow x = -7 \\ & |x-2| - 3 = -7 \end{aligned}$$

$$|x-2| = -4$$

ϕ .

$$\begin{aligned} & \overset{\text{---}}{\underset{\text{---}}{|x-2| = 10}} \\ & \swarrow \quad \searrow \\ & x-2 = 10 \qquad x-2 = -10 \\ & \textcircled{x = 12} \qquad \textcircled{x = -8} \end{aligned}$$

Solve both equations:

(i) $|x - 1| = |2x - 3|$

$$(x - 1) = \pm (2x - 3)$$

$$x - 1 = 2x - 3$$

$$2 = x$$

✓

$$x - 1 = -2x + 3$$

$$3x = 4$$

$$x = \frac{4}{3}$$

✓

(ii) $|x - 1| = 2x - 3$

$$(x - 1) = \pm (2x - 3)$$

$$x = 2, \frac{4}{3}$$

$x = 2$
only

Check:

$$|x - 1| = 2x - 3$$

$$x = 2$$

$$1 = 1 \quad \checkmark$$

$$x = \frac{4}{3}$$

$$\left| \frac{1}{3} \right| = 2 \left(\frac{4}{3} \right) - 3 = -\frac{1}{3} \quad \times$$

$$\frac{4}{3} \rightarrow$$



SACHIN SIR SPECIAL SUMMARY(S4)

For Modulus Equation:

Type-01: Ek side MOD, ek side constant: $|x| = 3$, $||x - 5| - 3| = 2$

Type-02: Both side MOD: $|x - 1| = |2x + 5|$

Type-03: EK side MOD, ek side variable: $|x - 1| = 2x + 5$

Type-04: Multiple MOD: $|x - 1| + 2|x - 4| = 8$.

Mod
हटाओ
+/-
लगाओ

(Cases बनाकर
MOD को हटाओ)

The number of real roots of the equation $(x+1)^2 + |x-5| = \frac{27}{4}$ is

$$x=5$$

C-I: $x \leq 5 \Rightarrow |x-5| = -(x-5) = 5-x$

C-II: $x \geq 5 \Rightarrow |x-5| = x-5$

2
fnf

$$(x+1)^2 + 5-x = \frac{27}{4}$$

$$x^2 + x + 6 = \frac{27}{4}$$

$$4x^2 + 4x - 3 = 0$$

$$4x^2 + 6x - 2x - 3 = 0$$

$$2x(2x+3) - 1(2x+3) = 0$$

$$x = \frac{1}{2}, -\frac{3}{2}$$

$$(x+1)^2 + (x-5) = \frac{27}{4}$$

$$x^2 + 2x + 1 + x - 5 = \frac{27}{4}$$

$$4x^2 + 12x - 16 - 27 = 0$$

$$4x^2 + 12x - 43 = 0$$

$$x = \frac{-\frac{3}{2} \pm \sqrt{4^2 \cdot 3^2 + 1(4)43}}{4 \times 2}$$

Rejected.

$$x = \frac{-3 \pm \sqrt{13}}{2}$$

$$x = \frac{-3 \pm 2\sqrt{13}}{2}$$

Question (Multiple MOD)

Solve for x : $|x-3| + 2|x-4| = 11$

Annotations: Above the 3 in $|x-3|$ is "-ve" with an arrow. Above the 4 in $|x-4|$ is "-ve" with an arrow. Below the 3 is a bracket labeled "3". Below the 4 is a bracket labeled "4".

$x = 0, \frac{22}{3}$ ✓

$x \leq 3$

$$-(x-3) + 2(-(x-4)) = 11$$

$$-x + 3 - 2x + 8 = 11$$

$$-3x = 11$$

$x = 0$ ✓

3

$3 \leq x \leq 4$

$$|x-3| + 2|x-4| = 11$$

$$x-3 + 2(-(x-4)) = 11$$

$$x-3 - 2x + 8 = 11$$

$$-x + 5 = 11$$

$x = -6$ ✗

4

$x \geq 4$

$$|x-3| + 2|x-4| = 11$$

$$x-3 + 2(x-4) = 11$$

$$3x - 11 = 11$$

$x = \frac{22}{3}$ ✓

The product of the roots of the equation $9x^2 - 18|x| + 5 = 0$ is

A $25/9$

B $5/27$

C $25/81$

D $5/9$



Question

Solution of inequality: $|x - 3| \leq |2x + 7|$.

A $x \in \left(-10, -\frac{4}{3}\right)$

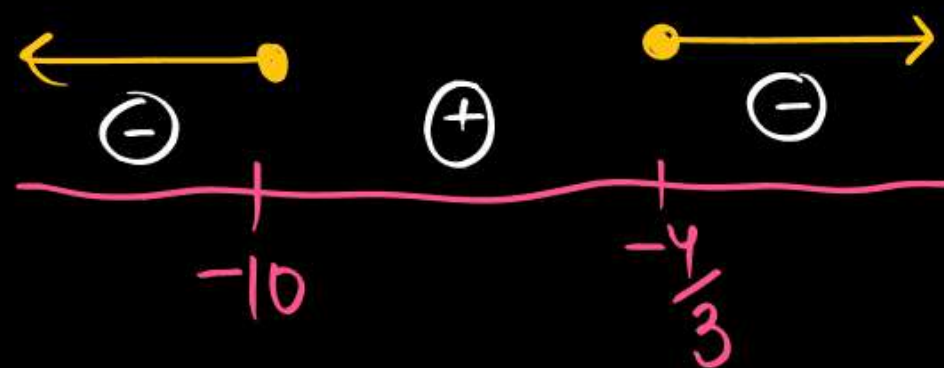
B $x \in (-\infty, -10] \cup \left[-\frac{4}{3}, \infty\right)$

C $x \in (-\infty, -10) \cup \left(-\frac{4}{3}, \infty\right)$

D $x \in \left[-10, -\frac{4}{3}\right]$

$$|x-3| - |2x+7| \leq 0$$

$$x = -10, -\frac{4}{3}$$



Rough

$$|x-3| - |2x+7| = 0$$

$$|x-3| = |2x+7|$$

$$x-3 = +(2x+7)$$

$$-10 = x \quad \checkmark$$

$$x-3 = -(2x+7)$$

$$x-3 = -2x-7$$

$$3x = -4$$

$$x = -\frac{4}{3} \quad \checkmark$$

Complete set of values of x satisfying: $|x + 3| - 2|x - 4| \geq 5$ is

How?

A $x \in \left(\frac{10}{3}, 6\right]$

B $x \in \left[\frac{10}{3}, 6\right] - \{4\}$

C $x \in \left[\frac{10}{3}, 6\right)$

D $x \in \left[\frac{10}{3}, 6\right]$

$$|x+3| - 2|x-4| - 5 \geq 0$$

Roots

$$|x+3| - 2|x-4| - 5 = 0$$

Question

Solve: $\frac{|x|-3}{|x+4|-1} \geq 0$

± 3

$$|x|-3=0$$

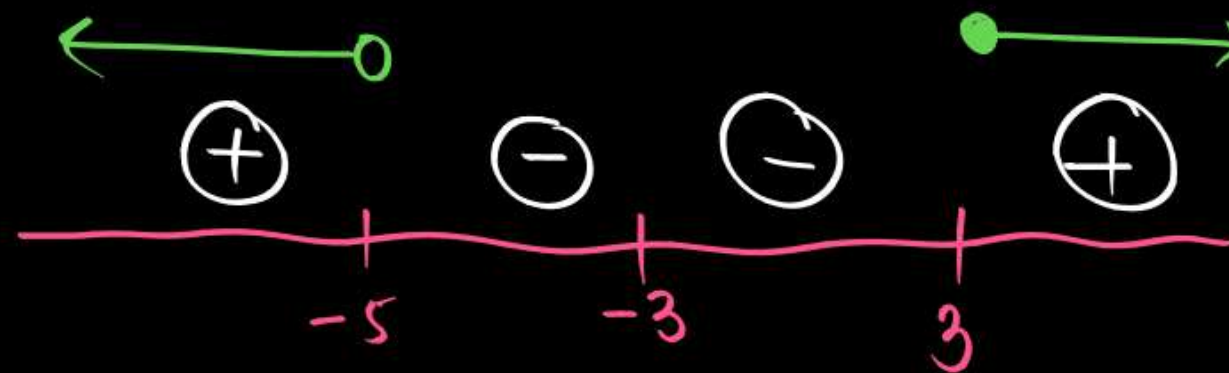
$$\hookrightarrow x = \pm 3$$

$\underline{\underline{-3, -5}}$

$$|x+4|=1$$

$$\hookrightarrow x+4 = 1, -1$$

$$x = -3, -5$$



$$x \in (-\infty, -5) \cup [3, \infty)$$

BONUS MOMENTS

Complete solution set of inequality $\frac{(x+2)(x+3)}{(x-2)(x-3)} \leq 1$ is

- A** $(-\infty, 0]$
- B** $(-\infty, 0] \cup (2, 3)$
- C** $[2, 3]$
- D** $(-\infty, 2) \cup (3, \infty)$

Complete solution set of inequality $\frac{(x+2)(x+3)}{(x-2)(x-3)} \leq 1$ is

- A $(-\infty, 0]$
- ☒ B $(-\infty, 0] \cup (2, 3)$
- C $[2, 3]$
- D $(-\infty, 2) \cup (3, \infty)$

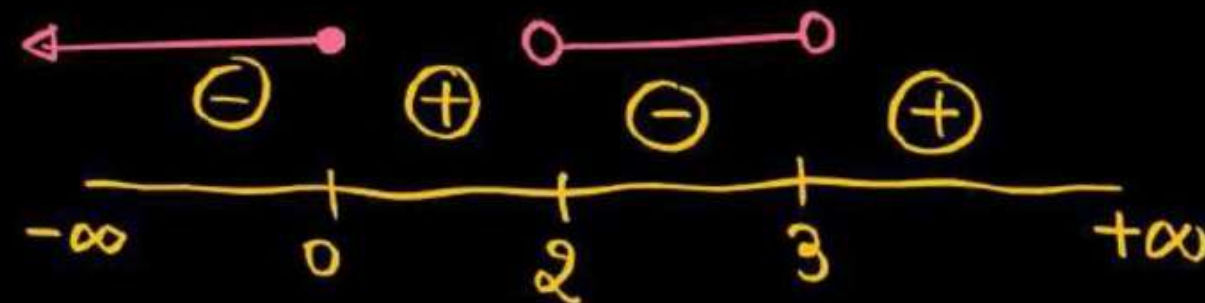
$$\frac{(x+2)(x+3)}{(x-2)(x-3)} - 1 \leq 0$$

$$\frac{x^2 + 5x + 6}{x^2 - 5x + 6} - 1 \leq 0$$

$$\frac{\cancel{x^2} + 5\cancel{x} + 6 - \cancel{x^2} + 5\cancel{x} - 6}{x^2 - 5x + 6} \leq 0$$

$$\frac{10x}{(x-2)(x-3)} \leq 0$$

$\swarrow \quad \searrow$
 2 3



Question



Solve : $\frac{x}{x-2} > \frac{1}{x+3}$

Question

Solve: $\frac{x}{x-2} > \frac{1}{x+3}$

$$\frac{x}{x-2} - \frac{1}{x+3} > 0$$

$$\frac{x(x+3) - (x-2)}{(x-2)(x+3)} > 0$$

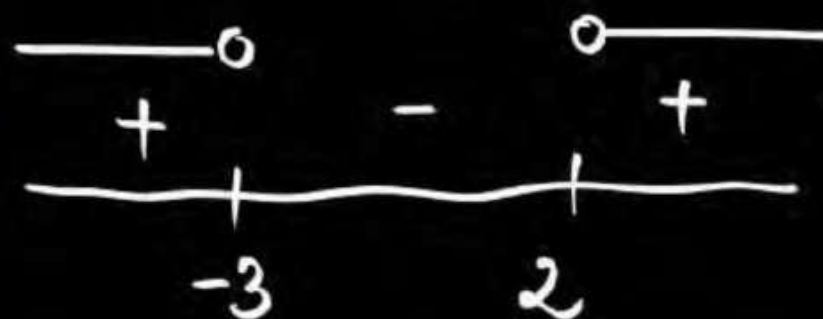
$$\frac{x^2 + 3x - x + 2}{(x-2)(x+3)} > 0$$

कोई Roots नहीं है ($\Delta < 0$)

[Ans. : $x \in (-\infty, -3) \cup (2, \infty)$]

$$\frac{x^2 + 2x + 2}{(x-2)(x+3)} > 0$$

$\downarrow \quad \quad \downarrow$
2 -3



Question



Solve for x : $|x - 1| - 2|x - 3| + 3|x + 1| = 10$

Question

"Multiple Mod"



Solve for x : $|x-1| - 2|x-3| + 3|x+1| = 10$

fnf $\Rightarrow$ $x = -9, \frac{7}{3}$
C-IV

$x \leq -1$

$-1 < x < 1$

$1 \leq x < 3$

$x \geq 3$

$\overset{-ve}{|x-1|} - 2\overset{-ve}{|x-3|} + 3\overset{-ve}{|x+1|} = 10$

$-(x-1) - 2(3-x) + 3(-x-1) = 10$

$-x+1-6+2x-3x-3=10$

$-2x-8=10$

$-2x=18$

$x=-9$ ✓

$\overset{-ve}{|x-1|} - 2\overset{-ve}{|x-3|} + 3\overset{+ve}{|x+1|} = 10$

$-(x-1) - 2(3-x) + 3(x+1) = 10$

$-x+1-6+2x+3x+3=10$

$4x-2=10$

$4x=12$
 $x=3$ ✗

$\overset{+ve}{|x-1|} - 2\overset{-ve}{|x-3|} + 3\overset{+ve}{|x+1|} = 10$

$(x-1) - 2(-(x-3)) + 3(x+1) = 10$

$x-1-2(3-x)+3x+3=10$

$x-1-6+2x+3x+3=10$

$6x=14$

$x=\frac{14}{6}=\frac{7}{3}$ ✓

$\overset{+ve}{|x-1|} - 2\overset{+ve}{|x-3|} + 3\overset{+ve}{|x+1|} = 10$

$x-1-2(x-3)+3(x+1)=10$

$x-1-2x+6+3x+3=10$

$2x+8=10$

$2x=2$

$x=1$ Ref. ✗

For $x \in R$, the number of real roots of the equation:

$$3x^2 - 4|x^2 - 1| + x - 1 = 0 \text{ is}$$

"Repeating"

For $x \in R$, the number of real roots of the equation:

$$3x^2 - 4|x^2 - 1| + x - 1 = 0$$
 is

Solⁿ $|x^2 - 1| = x^2 - 1, \quad x^2 - 1 \geq 0$
 $= -(x^2 - 1), \quad x^2 - 1 < 0$

fnf $\rightarrow$ 4

C-1: $x^2 - 1 \geq 0 \Rightarrow x \in (-\infty, -1] \cup [1, \infty)$

$$3x^2 - 4(x^2 - 1) + x - 1 = 0$$

$$3x^2 - 4x^2 + 4 + x - 1 = 0$$

$$-x^2 + x + 3 = 0$$

$$x = \frac{-1 \pm \sqrt{(1)^2 - 4(-1)(3)}}{2(-1)} = \frac{-1 \pm \sqrt{13}}{-2}$$

$\frac{1 - \sqrt{13}}{2} < -1$ (3. m)
 $\frac{1 + \sqrt{13}}{2} \rightarrow \frac{4. m}{2} = 2. m$



"Repeating"



For $x \in \mathbb{R}$, the number of real roots of the equation:

$$3x^2 - 4|x^2 - 1| + x - 1 = 0 \text{ is}$$

C-II:

$$x^2 - 1 < 0$$

$\Rightarrow$

$$x \in (-1, 1)$$

$$3x^2 - 4(-(x^2 - 1)) + x - 1 = 0$$

$$3x^2 + 4(x^2 - 1) + x - 1 = 0$$

$$3x^2 + 4x^2 - 4 + x - 1 = 0$$

$$7x^2 + x - 5 = 0$$

$$x = \frac{-1 \pm \sqrt{(1)^2 - 4(7)(-5)}}{2(7)}$$

$$= \frac{-1 \pm \sqrt{141}}{14}$$

$$\frac{-1 + \sqrt{141}}{14}$$



$$\frac{-1 - \sqrt{141}}{14}$$





Homework



Re-attempt Important Questions of Class.

Solve DPP

THANK YOU

